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## Logistic Regression

LR is a binary classification algo that predicts the prob. of a class (mostly binary 0/1).

Ex: Spam / Not spam.

$$y \in (0,1).$$

$$\hat{y} = P(y=1/x).$$

For  $k$  features  $x_1, x_2, x_3, x_4 \dots x_k$ .

$$x_i^0 = x_{i1} + x_{i2} + x_{i3} \dots x_{ik}.$$

$$w = [w_1, w_2, w_3, \dots, w_k].$$

$$z_i^0 = w_1 x_{i1} + w_2 x_{i2} + \dots + w_k x_{ik} + b$$

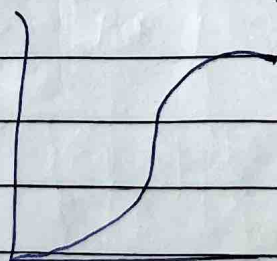
$$\hat{y}_i = \sigma(z_i^0) = \frac{1}{1 + e^{-(z_i^0)}}$$

$$y_i = \begin{cases} 1 & \text{if } \hat{y}_i \geq 0.5 \\ 0 & \text{if } \hat{y}_i < 0.5 \end{cases} \quad \text{Decision Rule}$$

Loss Fxn : (Binary Cross Entropy loss)

$$J(w, b) = -\frac{1}{m} \sum_{i=1}^m [y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i)]$$

PTO →





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Gradient wrt  $w_k$ :

$$\frac{\partial J}{\partial w_j^0} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij}^0$$

Gradient wrt to Bias:

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)$$

Update:

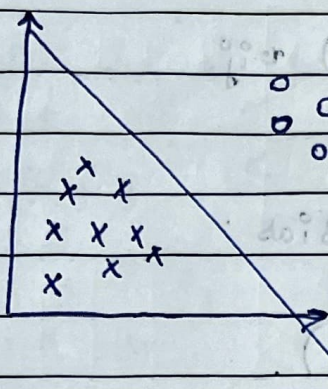
$$w_j^0 = w_j^0 - \alpha \frac{\partial J}{\partial w_j^0}$$

$$b = b - \alpha \frac{\partial J}{\partial b}$$



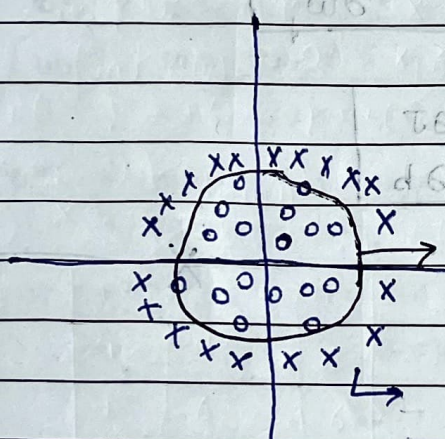
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## Decision Boundary



Decision Boundary  
is linear.

If Data is linearly ~~separable~~ separable  
we use the LR model  
→ linear Regression



Decision  
Boundary.

Not linearly separable  
Data.

$$f = w_1 x_1^2 + w_2 x_2^2 + b.$$

Let  $w_1 = 1$ ,  $w_2 = 1$ ,  $b = -2 \rightarrow$  After  
train.

$$g(w_1 x_1^2 + w_2 x_2^2 + b).$$

$$f = w_1 x_1^2 + w_2 x_2^2 + b.$$

$$x_1^2 + x_2^2 - 2 \rightarrow \text{For } f = 0.$$

$$x_1^2 + x_2^2 = 2$$

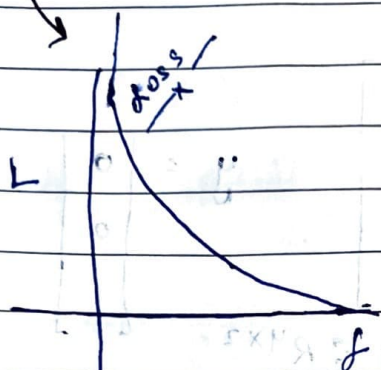
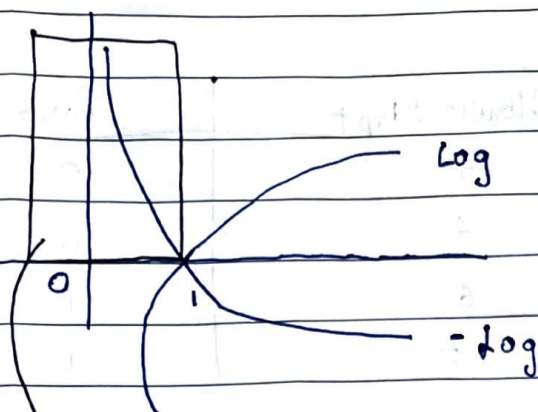
Circle

Eq. of the Decision  
Boundary



## Log Graph

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Code : logistic Reg from Stattech.  
Mathematic of Logistic Reg.  
TP / TN / FP / FN

• Precision

• Recall

F1 Score

Accuracy.

Confusion  
Matrix.

Ex: Dataset

| Hours Studied | Hours Slept | Pass |
|---------------|-------------|------|
| 2             | 5           | 0    |
| 4             | 6           | 0    |
| 6             | 6           | 1    |
| 8             | 7           | 1    |

We Define :

$$X = \begin{bmatrix} 2 & 5 \\ 4 & 6 \\ 6 & 6 \\ 8 & 7 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

No of Samples = 4  $X \in \mathbb{R}^{4 \times 2}$

No of features = 2  $x_i \in \mathbb{R}^{2 \times 1}$

Add Bias Column

$$X = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 6 \\ 1 & 6 & 6 \\ 1 & 8 & 7 \end{bmatrix}$$

Shape becomes  $X \in \mathbb{R}^{4 \times 3}$

Each row becomes  $x_i = [1, x_{i1}, x_{i2}]$

Eg,

$$x_1 = [1, 2, 5]$$

$$x_2 = [1, 4, 6]$$

Initialize Weight

$$w = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

$$w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



## Linear Combination :

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Compute :

$$z = Xw$$

$$z = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 6 \\ 1 & 6 & 7 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow w^T x_i + b$$

Each prediction before sigmoid

$$z_i = w^T x_i$$

Sigmoid  $\frac{z}{x}$

$$\hat{y} = \sigma(z)$$

Since,

$$z = 0$$

$$\sigma(0) = 0.5$$

$$\hat{y} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Cost  $\frac{z}{x}$

$$J(w) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

where  $m = 4$

$$y_i = 0 \quad \hat{y}_i = 0.5 \quad \text{for All points}$$

$$J = \frac{1}{4} (0.693 + 0.693 + 0.693 + 0.693)$$

$$J = 0.693$$



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# # Gradient Computation.

$$\nabla J = \frac{1}{m} X^T (\hat{y} - y)$$

$$\hat{y} - y = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 8 \\ 5 & 6 & 6 & 7 \end{bmatrix}$$

$$X^T (\hat{y} - y)$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 8 \\ 5 & 6 & 6 & 7 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1(0.5) + 1(0.5) + 1(-0.5) + 1(-0.5) \\ 2(0.5) + 4(0.5) + 6(-0.5) + 8(-0.5) \\ 5(0.5) + 6(0.5) + 6(-0.5) + 7(-0.5) \end{bmatrix}$$

$$\Rightarrow \nabla J = \begin{bmatrix} 0 \\ -4 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ -1 \\ -0.25 \end{bmatrix}$$

Update rule

$$w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 0 \\ -1 \\ -0.25 \end{bmatrix}$$

$$\Rightarrow w = \begin{bmatrix} 0 \\ 0.1 \\ 0.025 \end{bmatrix}$$

Model

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## Confusion Matrix

A confusion matrix is a table used to evaluate a classification model by comparing:

- Actual labels
- Predicted labels

Structure:

|                 | Pred Positive | Predicted Negative |
|-----------------|---------------|--------------------|
| Actual Positive | TP            | FN                 |
| Actual Negative | FP            | TN                 |

## Accuracy

Accuracy =  $\frac{\text{Correct Prediction}}{\text{Total prediction}}$

Mathematically

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Out of the predicted positive how many were actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision →



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## Recall (Sensitivity)

Out of all actual positives, how many did the model detect?

$$\text{Recall} = \frac{TP}{TP + FN}$$

## F1 Score :

F1 Score combines precision & recall into one metric.

It is the harmonic mean of precision & recall.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$